MERAJ ANSARI

Full-Stack Engineer | Backend Systems & SaaS Architecture | Distributed Workers & Automation

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 2.5 years of production experience designing and operating distributed worker architectures, scalable SaaS backends, and high-performance automation workflows at PerfectlyOkay Solutions. Deeply hands-on with Node.js, TypeScript, MongoDB, Next.js, and browser engines (Puppeteer/Playwright). Known for taking end-to-end technical ownership: architecting lease-based worker queues with zero database exposure, engineering sub-second quick-commerce scraping pipelines, optimizing database read paths by 99.8%, and applying strict defensive patterns against system failures.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3
Backend Systems	Node.js, Express.js, REST APIs, Distributed Job Queues, WebSocket / Socket.IO, SSE, Cron Scheduling, Transactional Outbox
Databases	MongoDB (Aggregation Pipelines, Index Optimization, Partitioned Collections), MySQL (Schema Design, Migration)
Automation & Scraping	Puppeteer, Playwright, Camoufox, Remote Worker Pools, Off-Screen Headful Execution, Session Stores, Proxy Routing
AI & Generative AI	OpenAI, Anthropic Claude & Google Gemini APIs, Multi-Model Routing, Post-Generation Hallucination Validators
Security & DevOps	JWT, OAuth 2.0, NextAuth, RBAC, TOTP 2FA, AES-256-GCM, SSRF Defense, Linux (Systemd & Cgroups), Git, Docker

PROFESSIONAL EXPERIENCE

Full-Stack Developer · PerfectlyOkay Solutions Pvt. Ltd. **Apr 2024 – Present**
Ulhasnagar, Thane, Maharashtra *Deal Engine, Deal Platform & Internal HR Portal*

- **Distributed Scraping Worker Architecture:** Designed and implemented the offer-scrape worker system; decoupled browser execution from the central database, enabling remote/local workers to claim jobs over HTTPS via atomic 2-minute leases, 30s heartbeats, and automatic server fallback when offline. Wrapped Puppeteer with injected off-screen window parking (`--window-position`) to run headful Chrome without disturbing user desktops while evading bot-detection.
- **Quick-Commerce Price Aggregation Engine:** Built real-time price & product search across 6 platforms (Blinkit, Zepto, JioMart, DMart, Flipkart Minutes, Amazon Now); engineered an adaptive browser pool with session snapshot persistence that slashed warm query latency to **0.17s – 0.56s** with 4 parallel platform fetches, down from 5–14s in cold setups.
- **Automated Coupon Verification Pipeline:** Integrated 26 affiliate network pipelines and built automated coupon validation for WooCommerce (Store API first, proxy fallback, stealth Chrome) and extended Shopify checks; built live WebSocket verification streaming, masked coupon code detection, and platform detection crons.
- **Database Migration & Tenant Partitioning:** Migrated legacy system from MySQL (24 tables) to MongoDB using Node.js and Express; redesigned architecture with 13 physically partitioned collection types per project to guarantee complete data isolation; eliminated a paging data-loss bug by replacing skip/limit with keyset pagination.
- **Query Performance & Synchronization:** Rewrote analytics queries on 2.4M+ documents—slashed dashboard scans from **307 to 6**, examined documents from **45.5M to 0.56M**, and cut activity feed reads from **21.9M to 50.6k (99.8% drop)**; synchronized Deal Engine and Platform via a transactional outbox with 15s ack timeouts, dead-letter queues, and atomic withdrawal claims preventing double-spends.
- **AI Assistants & Enterprise Security:** Integrated OpenAI, Claude, and Gemini to power a grounded deal assistant with 4 post-generation hallucination validators, lifting prompt check passes from 22/43 to 43/43; built the HR portal's salary history module with AES-256-GCM encryption at rest and patched stored XSS and unauthorized blog APIs.

Web Developer Intern · Learning Beyond Your Horizon (CBSE) **Jan 2024 – Apr 2024**
Remote *HTML5, CSS3, JavaScript*

- Developed the official homepage and modular responsive sections, optimizing page layout and cross-browser rendering.

Freelance Web Developer · Local News Channel **Aug 2023 – Nov 2023**
Maharashtra, India *React.js, REST APIs*

- Engineered dynamic React.js frontend views integrated with backend REST APIs for real-time article publishing and categorization.

FLAGSHIP ENGINEERING PROJECTS

BrandSecurer — Multi-Tenant Brand Protection SaaS **Next.js, TypeScript, Node.js, MongoDB, Puppeteer, Paddle**

- Architected an ad violation tracker monitoring paid and organic search across **8 search engines in 42 selectable countries**; built a 20-hop Puppeteer redirect tracer identifying affiliate IDs across 34 networks with SSRF validation on every redirect hop.
- Implemented a circuit breaker (3 failures trigger 90s probe) and failover routing across 6 SERP providers; integrated Paddle billing with signature-verified idempotent webhooks (14 event types), failed-event replay, and 15-minute reconciliation.

Backupper — Server Backup, Restore & Health Monitoring Platform **Next.js, Node.js, MongoDB, MySQL, SSH2 / SFTP, Systemd / Cgroups**

- Built an SSH-driven backup engine for remote Linux servers (files, MongoDB, MySQL) enforcing systemd cgroup CPU and memory limits inside scopes; eliminated ~15 GB RAM overhead on 9 GB exports by streaming NDJSON.
- Implemented staged 30-minute preview restores with SHA-256 diffing and automatic rollback; built telemetry monitoring featuring 7-day median baseline spike detection, least-squares disk-fill forecasting up to 30 days, and SSE alerts.

EDUCATION

Bachelor of Engineering (B.E.) in Information Technology — M.H. Saboo Siddik College of Engineering (Mumbai University) 2019 – 2023

Higher Secondary Certificate (HSC) in Science — The Scholar's High School and Junior College 2017 – 2019